

NADIA GUL

Machine Learning Engineer | Deep Learning · MLOps · Data Analysis

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Summary

Computer Science graduate (MScS) with hands-on expertise in machine learning, deep learning, and end-to-end MLOps. Experienced in fine-tuning multimodal and vision models, applying Explainable AI (Grad-CAM, LIME, SHAP), and shipping production-grade ML pipelines with CI/CD, containerization, and automated testing. Strong background in technical writing and documentation, with a proven ability to communicate complex ML concepts clearly to diverse audiences.

Experience

Artificial Intelligence Intern | Genesys Research Lab Jun 2026 – Aug 2026
Summer Internship Program · Mentor: Dr. Muhammad Mateen Yaqoob

- Built an **end-to-end Explainable Image Captioning system** using BLIP on the Flickr8k dataset, covering data preparation, fine-tuning, and deployment.
- Fine-tuned BLIP for image captioning, achieving a **29% improvement in METEOR score** over the baseline.
- Integrated **Explainable AI (Grad-CAM, LIME)** to visualize and interpret model predictions.
- Engineered a full **MLOps pipeline** with MLflow, DVC, FastAPI, Docker, and GitHub Actions CI/CD.
- Implemented **20 pytest unit tests** to improve reliability and maintainability, and deployed a live demo on **Hugging Face Spaces**.

Teaching Assistant – Data Visualization | FAST-NUCES 2026
Islamabad, Pakistan

- Evaluated student assignments and projects using structured rubrics to ensure consistent, fair grading.
- Designed assignments and coursework aligned with learning objectives and practical data-visualization skills.
- Provided feedback and clear explanations to help students improve technical understanding and presentation of data.

Web Developer | Vantage Droid Private Limited Oct 2023 – Feb 2024

- Developed a student login system for authentication and user management.
- Designed an admin panel using SQL for database operations.
- Built a responsive UI using HTML, CSS, and JavaScript.

Education

Master of Science in Computer Science | FAST-NUCES 2024 – 2026
CGPA: 3.26 · Thesis (In Progress): Resource-Aware Federated Learning for IoT — Efficient Client Selection & Lightweight Models

- Designing resource-aware client selection based on computation, energy, and communication constraints.
- Developing a lightweight FL framework for heterogeneous IoT environments with adaptive aggregation (synchronous, asynchronous, hybrid).
- Simulating non-IID data distributions and evaluating accuracy, convergence, communication cost, and fairness.

Bachelor of Information Technology | University of Sargodha Graduated
CGPA: 3.29 · Final Year Project: Badminton Ontology — Semantic Information Retrieval

- Developed an ontology-based semantic information-retrieval system for the sports domain using the Neon Methodology.
- Designed a structured framework representing player skills, strategies, and equipment for intelligent coaching systems.

Technical Skills

Machine Learning	KNN, Decision Trees, Random Forest, Gradient Boosting, SVM, Cross-Validation
Deep Learning	BLIP, Transformers, CNNs, MLP, wav2vec2, Adversarial Training (GRL)
Explainable AI	Grad-CAM, LIME, SHAP
MLOps & Deployment	MLflow, DVC, FastAPI, Docker, GitHub Actions (CI/CD), pytest, Hugging Face Spaces
Computer Vision & Multimodal	Image Captioning, Feature Extraction, Speech Processing (Librosa, Parselmouth/Praat)
Programming & Frameworks	Python, PyTorch, Scikit-learn
Data Analysis & Visualization	Tableau, Excel, Data Preprocessing, Feature Engineering
Cloud	AWS (Cloud Security, Builder Labs)
Technical Writing	ML Reports, Research Writing, LaTeX, MS Office (Word, Excel, PowerPoint)

Projects

Explainable Image Captioning with BLIP | [Genesys Research Lab](#)

2026

PyTorch · BLIP · Grad-CAM · LIME · MLflow · DVC · FastAPI · Docker · GitHub Actions · Hugging Face

- Fine-tuned BLIP on the Flickr8k dataset, improving the METEOR score by 29% over baseline.
- Applied Grad-CAM and LIME for model interpretability and prediction visualization.
- Delivered a reproducible, tested, and containerized pipeline deployed live on Hugging Face Spaces.

VSAA-AC: Speaker-Invariant Accent Classification

2025

PyTorch · MFCCs · wav2vec2 · Gradient Reversal Learning

- Developed a speaker-invariant English accent classifier using MFCCs, vowel formants (F1/F2), and Gradient Reversal Learning.
- Built and evaluated SVM, MLP, wav2vec2, and adversarial models on the Speech Accent Archive dataset.
- Improved unseen-speaker generalization via adversarial training; validated with ablation studies, Macro-F1, and McNemar testing.

Breast Cancer Prediction (KNN)

2025

- Achieved **96.3% accuracy and 98% recall** on the Breast Cancer Wisconsin dataset (569 samples, 30 features).
- Applied PCA and feature pruning to reduce dimensionality by up to 40% with minimal performance loss; compared KNN vs SVM trade-offs.

Heart Disease Prediction (Ensemble Methods)

2025

- Built Decision Tree, Random Forest, and Gradient Boosting models reaching ~81.7% accuracy with strong recall.
- Addressed overfitting through pruning and hyperparameter tuning; proposed SHAP-based explainability and probability calibration.

Systematic Literature Review — AI Clinical Trial Matching

2025

- Synthesized NLP, ML, and hybrid approaches across IEEE, PubMed, Springer, and Scopus, evaluating MIMIC-III, N2C2, and ClinicalTrials.gov.
- Identified research gaps and proposed future directions in Explainable AI for healthcare systems.

Interactive Data Storytelling — FIFA Dataset | [Tableau](#)

2024

- Analyzed a 17,000+ player dataset to uncover trends in ratings, age, and nationality through interactive dashboards with KPIs and filters.

Certifications

- **Prompt Engineering for ChatGPT** — Vanderbilt University (Coursera), 2026
- **AWS Academy Graduate – Cloud Security Foundations** — Training Badge, 2025
- **AWS Academy Graduate – Cloud Security Builder** — Training Badge, 2025
- **Introduction to AWS Cloud: Builder Labs Learning Plan** — AWS Training & Certification, 2025
- **Programming for Everybody (Getting Started with Python)** — University of Michigan (Coursera), 2025
- **Write Professional Emails in English** — Georgia Institute of Technology (Coursera), 2025
- **The Complete Web Development Bootcamp** — Udemy, 2023

Languages

English • Urdu • Punjabi • Saraiki

Volunteer Experience

Member | [Information Technology Society, University of Sargodha](#)

- Participated in the Information Technology Society, contributing to student tech activities and events.